

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

(i) What is the V-Shaped Model also known
as?

✓ a) Verification and Validation Model

b) Spiral Model

c) Iterative Model

d) Agile Model

(ii) A key limitation of the V-Shaped Model is :

✓ a) Lack of flexibility to changes during the development process

b) High cost of implementation

c) It requires complex tools and technologies

d) It is not suitable for small projects

(iii) Which testing strategy involves testing each module individually before integrating them ?

a) System Testing

b) Integration Testing

✓c) Unit Testing

d) Regression Testing

(iv) Which of the following is NOT typically included in a test strategy?

a) Testing tools to be used

b) Testing schedule

✓c) Code implementation details

d) Scope of testing

(v) Validation testing is primarily concerned with:

a) Finding and fixing defects

✓b) Ensuring the software meets the user's needs

c) Ensuring all individual units function correctly

d) Testing the software's security

(vi) What is the main goal of acceptance testing?

a) To ensure the software meets the requirements of the development team

b) To ensure the software meets the user's needs and requirements

c) To test the software's performance under load

d) To find as many bugs as possible

(vii) System testing typically involves testing:

a) Individual units of code

b) The complete and integrated software system

c) The software's installation procedures

d) The software's documentation

(viii) Which term is used to describe a testing technique that focuses on the internal structure of the software?

- a) Black-box Testing
- ✓ b) White-box Testing
- c) Grey-box Testing
- d) Beta Testing

(ix) In Basis Path Testing, the number of independent paths to be tested is determined by:

- a) The number of test cases available
- ✓ b) The cyclomatic complexity of the program
- c) The number of variables in the program
- d) The total lines of code

(x) Cyclomatic complexity is a metric used to measure:

- a) The number of test cases needed
- ✓ b) The complexity of the code

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- c) The number of executable statements
 - d) The number of defects in the code

Section-B

Short answer type questions

Answer any **four** questions of the following :

5×4=20

- 2. What is Black Box Testing, and what are its primary objectives?
- 3. Differentiate between black box testing and white box testing.
- ✓ 4. Describe briefly the process of executing Boundary Value Analysis in software testing.
- ✓ 5. How are verification and validation defined in software testing, and what roles do they play?
- 6. Why is software testing important?
- ✓ 7. Write the different between functional and non-functional testing.

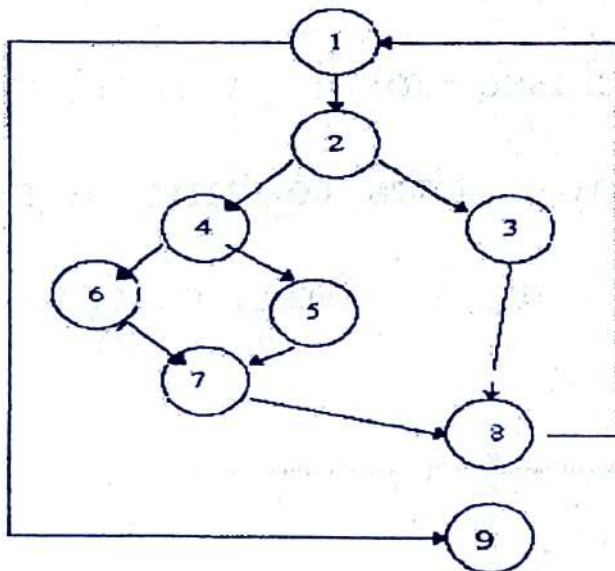
Section-C

Long answer type questions

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

8. Explain V-Shaped software lifecycle model with suitable figure.
9. Answer the following questions based on the figure given below:



- (a) Calculate the cyclomatic complexity using the predicate node.

- (b) Calculate the cyclomatic complexity using the regions.
 - (c) Calculate the cyclomatic complexity using edges and nodes.
 - (d) Show how many independent paths are exist.
10. Explain the different types of coverage techniques.
11. Explain the steps involved in identifying equivalence classes for a given set of input data. How does this technique help in reducing the number of test cases?

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